温州肯恩大学
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MGS 3101 W03 Foundations of Business Analytics

Simple Linear Regression II

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Review

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$y = \beta_0 + \beta_1 x + \varepsilon \qquad \hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{Cov(x,y)}{Var(x)}$$

- p-value
  - Probability of observing a test statistic as extreme as the sample result ***if*** the null hypothesis is true.
  - For a upper-tailed test: $P(\bar{X} \geq \bar{x}) = P\left(t \geq \frac{\bar{x} - \mu}{s/\sqrt{n}}\right)$
  - Measure the fit between sample data and H_0.
  - Interpretation: A smaller p-value $\rightarrow$ sample data is less consistent with H_0 $\rightarrow$ reject H_0

- Simple Linear Regression
  - The estimated coefficients $(\hat{\beta}_0, \hat{\beta}_1)$ come from sample data and estimate the population parameters (β_0, β_1).
  - OLS estimator: choose $(\hat{\beta}_0, \hat{\beta}_1)$ that minimize the sum of squared residuals $\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$
  - Assess model fit
    - Coefficient of determination $R^2 = \frac{SSR}{SST} = \frac{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$, measure the proportion of variation in y explained by the regression model.

Today's Outline

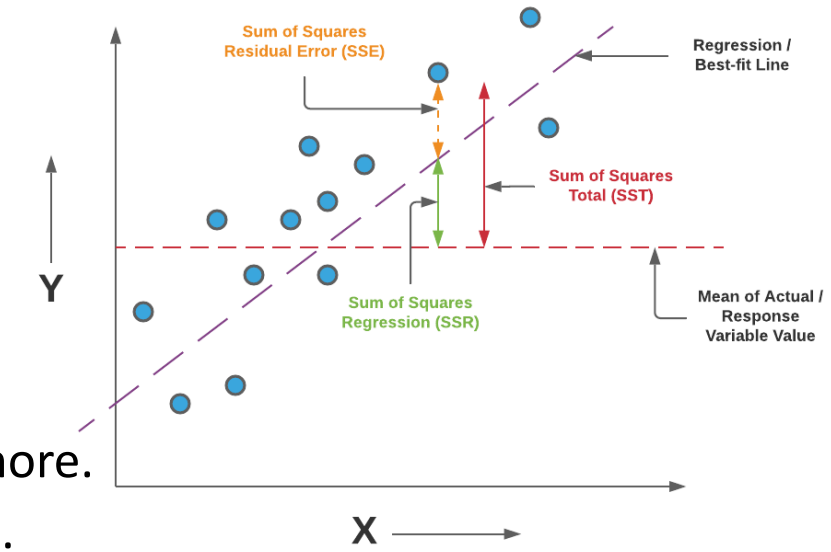
- How Good Is the Model?
  - Coefficient of determination: R^2 and adjusted R^2
- How Good Are the OLS Estimator ($\hat{\beta}_0$, $\hat{\beta}_1$)?
  - Unbiasedness of OLS estimator
  - Gauss-Markov Assumptions → OLS estimator is **BLUE** (Best Linear Unbiased Estimator)
- Inference for Regression Coefficients
  - Sampling distribution of $\hat{\beta}_0$, $\hat{\beta}_1$
  - Hypothesis testing of $\hat{\beta}_0$, $\hat{\beta}_1$
- Hands-On in Python
  - Running simple linear regression
  - Interpreting the regression results

Assessing Model Fit $\boldsymbol{R^2}$

- R^2 measures how much of the variation in the dependent variable y is explained by the regression model.

$$R^2 = \frac{\text{Explained Variation}}{\text{Total Variation}} = \frac{SSR}{SST} = \frac{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} = 1 - \frac{SSE}{SST} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

- Properties
  - $0 \leq R^2 \leq 1$
  - $R^2 = 0$: model explains none of the variation.
  - $R^2 = 1$: model explains all variation (perfect fit).
- Intuition
  - Think of points scattered around the regression line.
  - Higher R^2 → points are closer to the line → model explains more.
  - Lower R^2 → points are more scattered → model explains less.



Source

Properties of OLS Estimator

- Unbiasedness
  - $E[\hat{\beta}] = \beta$ under the zero-conditional mean assumption.
  - On average, OLS estimator hits the true parameter.
- Consistency
  - $\hat{\beta} \xrightarrow{p} \beta$ as $n \to \infty$
  - With larger sample size, OLS estimator gets arbitrarily close to the true value.
- Efficiency (BLUE)
  - Under the full Gauss-Markov assumptions (homoscedasticity & no autocorrelation), OLS estimator has the **smallest variance** <u>among all linear unbiased estimators</u>.

Not Required

Gauss-Markov Assumptions

A7. Normality (Optional, not Gauss-Markov)

$$\varepsilon_i \sim N(0, \sigma^2)$$

Not needed for BLUE, but need for: exact **t-tests**, exact **confidence intervals**, **F-test**

A1. Linearity in Parameters

The true model is linear in β: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$

A2. Random Sampling

(x_i, y_i) are drawn from the population as i.i.d observations.

A3. No Perfect Multicollinearity

Regressors must not be perfectly linearly related. In simple regression, x must vary.

A4. Zero Conditional Mean (Exogeneity)

$E[\varepsilon_i | x_i] = 0$, the error term is uncorrelated with x → ensure unbiasedness

A5. Homoscedasticity

$Var(\varepsilon_i | x_i) = \sigma^2$, constant variance of the error term → necessary for OLS estimator to be efficient (minimum variance)

A6. No Autocorrelation

$Cov(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$, Error are independent across observations.

Not Required

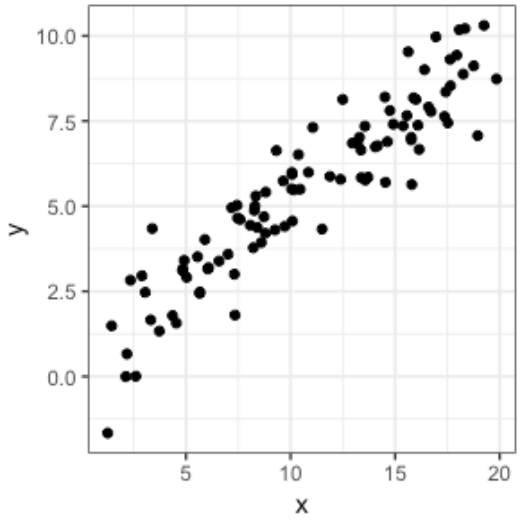
Assumptions Check

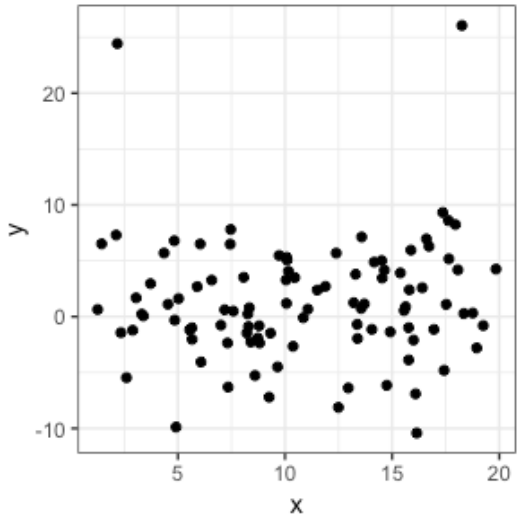
Assumption	Way to Check
A1. Linear relationship between DV and IV	Examine X-Y scatter plot, relationship should look linear.
A3. No perfect multicollinearity (multiple regression only)	Check correlation matrix, VIF (Variance Inflation Factor): VIF > 10 indicates multicollinearity.
No outliners	Examine X-Y scatterplot/histograms for outliners
A4. Zero conditional mean **A5.** Homoscedasticity	Plot residuals vs fitted value ($\hat{y}$): residuals should be random, the spread should be roughly constant (no funnel shape).
A6. No autocorrelation	Plot residuals vs observation order (time); should show no patterns. For time series: run Durbin–Watson test.
A7. Normally-distributed errors	Check histogram or Q–Q plot of residuals; residuals should follow a straight line in Q–Q plot.

Not Required

Assumptions Check

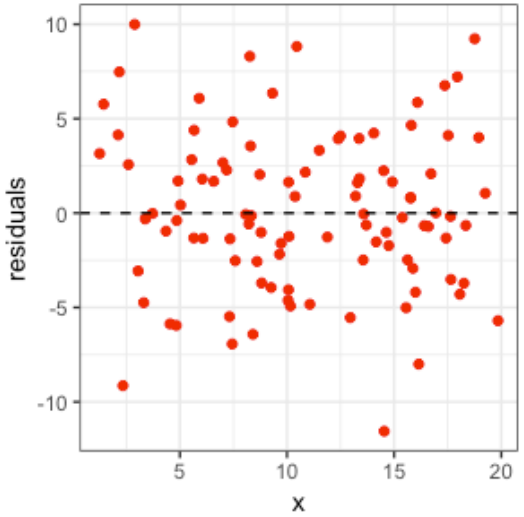
Linear
No outliners



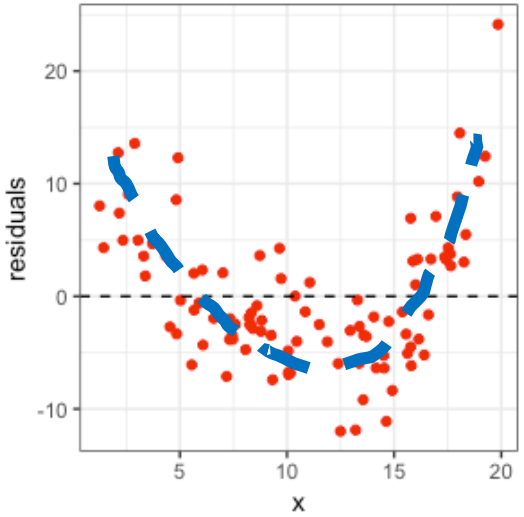


Nonlinear
With outliners

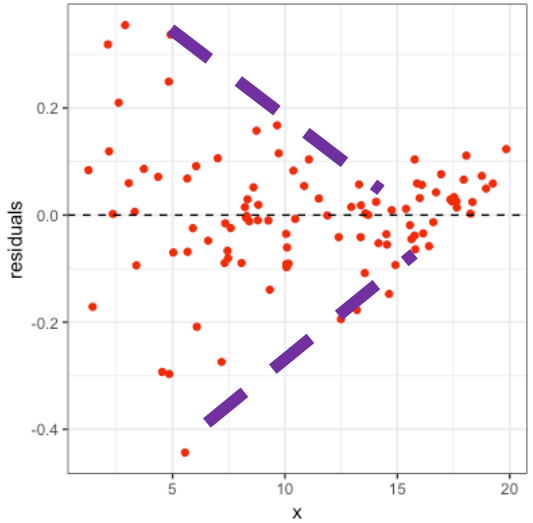
$E[\epsilon_i|x_i]$



Ideal



Non-linearity

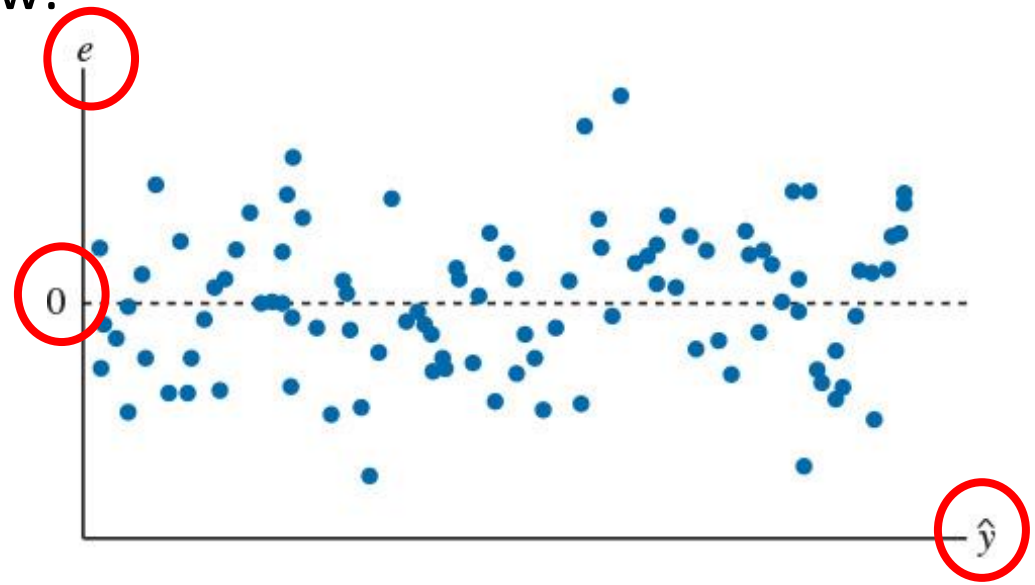


Heteroscedasticity

Not Required

Residual Plot: Checking Error Term Assumptions

- A simple scatter chart of the residuals is one of the most effective ways to diagnose violations of the error term ε_i assumptions.
- The example on the right shows a random error pattern in a scatter plot of residuals vs. the predicted values of the dependent variable ($\hat{y}$).
- For valid inference, the residual plot should show:
  - Residuals centered around zero (no systematic pattern, zero conditional mean)
  - Constant spread across all fitted values (homoscedasticity)
  - Symmetric distribution of residuals, with values near zero occurring most frequently (normality)

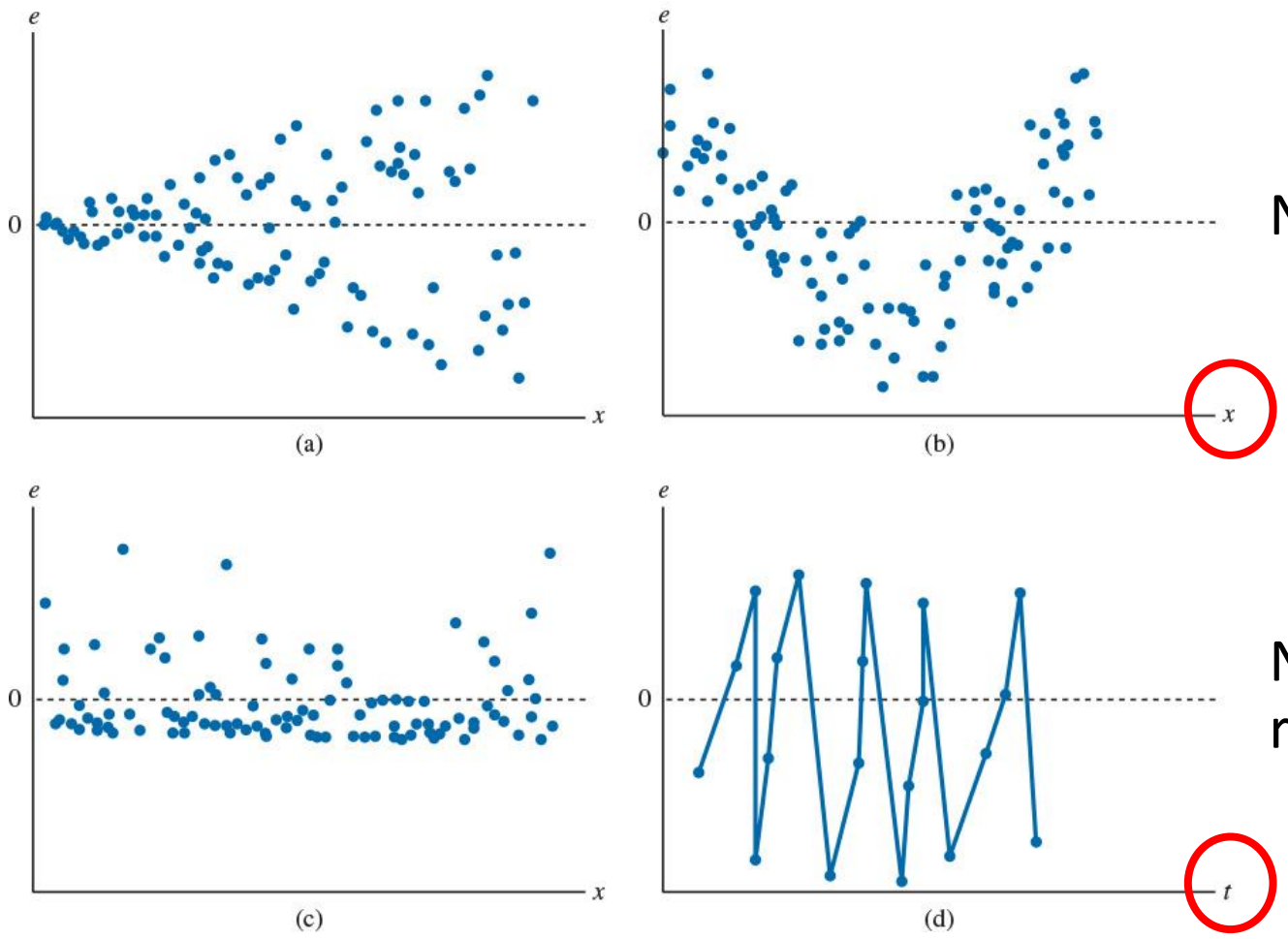


Not Required

Assumptions Check – Common Error Term Violations

Non-constant spread

Nonlinear pattern



Non-normal residuals

Non-independent residuals

No Need to Memorize.

Inferences for Regression Coefficients

Sampling Distribution of OLS Estimator

- $\hat{\beta}_0$ and $\hat{\beta}_1$ are random variables.
- Under the OLS assumptions
  - They are unbiased estimators

$$E[\hat{\beta}_0] = \beta_0 \qquad E[\hat{\beta}_1] = \beta_1$$

  - They have sampling distributions

$$\hat{\beta}_j \sim N(\beta_j, Var(\hat{\beta}_j)) \quad \text{if errors are normal}$$

  - Their variances shrink as sample size increases.

Confidence Interval (CI)

- A $(1-\alpha) \times 100\%$ CI for β_j:

$$\hat{\beta}_j \pm t_{\alpha/2,n-2} SE(\hat{\beta}_j)$$

- An **estimator** is a random variable: a rule or formula used to estimate a population parameter (e.g., $\hat{\beta}_1$, $\bar{x}$).
- An **estimate** is the numerical value obtained when the estimator is applied to a specific sample.
- Different samples → different estimates from the same estimator.

No Need to Memorize.

Hypothesis Testing for Regression Coefficients

Purpose

- Recall: Use sample data to make inference about the population parameters.
- Determine whether a regression coefficient β_j is statistically significant.
- Test whether the population parameter $\beta_j \neq 0$.
- Assess whether the IV has a meaningful impact on the DV

$$H_0: \beta_j = 0 \qquad\qquad H_1: \beta_j \neq 0$$

Test Statistic

$$t = \frac{\hat{\beta}_j - \beta_{j,0}}{SE(\hat{\beta}_j)} \sim t_{n-k}$$

where k is the number of parameters

Decision Rule

- Compute p-value from the t-statistic.
- Reject H_0 is $p < \alpha$.
- Otherwise, fail to reject H_0.

Hands-On in Python

Refer to the Jupyter notebook

demo_simple_linear_regression.ipynb

for the full demonstration.

Summary

- Model Fit
  - Define and interpret R^2
  - Assess how well the regression line explains variation in y
- Properties of OLS Estimators
  - Unbiasedness, consistency, and efficiency (BLUE)
  - The Gauss–Markov assumptions
- Inference for Regression Coefficients
  - Sampling distributions of $\hat{\beta}_0$ and $\hat{\beta}_1$
  - Standard errors, confidence intervals, and t-tests
  - Hypothesis testing: H_0: $\beta_j = 0$
- Checking OLS Assumptions
  - Use residual plots to assess error term ε assumption
- Hands-On in Python
  - Generate a population and random samples
  - Run simple linear regression and interpret the output
  - Simulate the sampling distribution of $\hat{\beta}_1$